MAE 3127: Introduction to Engineering Ethics

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On deliverables:

- **Deadline for submitting Ethics Poster Draft** is 11/23.
- **Deadline for submitting Finalized Ethics Poster Draft** is 12/01.
- **Ethics Poster Presentations:** 12/03
- Utilize office hours.
- Check into Ed discussion board: https://edstem.org/us/courses/83318/discussion

Introduction to Engineering Ethics

Slide courtesy of Prof. Snyder

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What Is Meant by “Engineering Ethics”?*

- **Engineering is Managing the Unknown:**
  - One source of ethical issues in engineering is **lack of knowledge**.
  - Engineering is about creating new devices and products—how well will they work under all conditions? How will they affect people?

- **Personal vs. Professional Ethics:**
  - **Personal Ethics** deals with how we treat others in our day-to-day lives.
  - **Professional Ethics** often involves choices on an organizational level rather than a personal level.

- **Ethics and the Law:**
  - Many things that are legal could be considered unethical (designing a process that releases a known toxic, but unregulated, substance is likely unethical, although it is legal).
  - Usually laws lag technology developments (e.g. unmanned aerial systems or cell phone encryption technology—Apple vs. FBI…).
  - **“Minimally safe” if you follow the requirements of applicable laws.**

*Fleddermann, Engineering Ethics, 4th Edition, Pearson, 2012.

What Is Meant by “Engineering Ethics”? (cont)

- **Ethics Problems are Like Design Problems:**
  - **Problems are more open ended** and are not as susceptible to formulaic answers like most engineering problems.
  - Need to apply a large body of knowledge and use analytical skills to determine possible solutions. **Rarely have a universal or fully correct answer.**
  - Requires detailed analysis, determination of specifications, often many “correct” or “acceptable” solutions. Approach Ethics like a Design Problem.
- **Best Understood Through Example (Case Studies):**
  - Point Paper in MAE 2131
  - Group Project in MAE 3127
- **Professional Society Guidance**

Professional Society Guidance*

- **American Society of Mechanical Engineers (ASME) has a *Code of Ethics of Engineers*.**
  - Similar code in other professional societies like American Society of Civil Engineers (ASCE), Association for Computing Machinery (ACM),…,

- ASME has both **Fundamental Principles** and **Fundamental Canons**.

- **ASME Fundamental Principles:**

  Engineers **uphold and advance the integrity, honor and dignity of the engineering profession** by:

  I. using their knowledge and skill for the enhancement of human welfare;

  II. being honest and impartial, and serving with fidelity their clients (including their employers) and the public; and

  III. striving to increase the competence and prestige of the engineering profession.

*https://www.asme.org/about-asme/get-involved/advocacy-government-relations/ethics-in-engineering

Professional Society Guidance (cont)

ASME Fundamental Canons:

1. Engineers shall **hold paramount the safety, health and welfare of the public** in the performance of their professional duties.

2. Engineers shall **perform services only in the areas of their competence**; they shall build their **professional reputation on the merit of their services** and **shall not compete unfairly** with others.

3. Engineers shall **continue their professional development throughout their careers** and shall provide opportunities for the **professional and ethical development of those engineers under their supervision**.

4. Engineers **shall act** in professional matters for each employer or client **as faithful agents or trustees**, and **shall avoid conflicts of interest** or the appearance of conflicts of interest.

Professional Society Guidance (cont)

ASME Fundamental Canons:

5. Engineers shall **respect the proprietary information and intellectual property rights** of others, including charitable organizations and professional societies in the engineering field.

6. Engineers shall **associate only with reputable persons or organizations**.

7. Engineers **shall issue public statements only in an objective and truthful manner** and shall avoid any conduct which brings discredit upon the profession.

8. Engineers shall **consider environmental impact and sustainable development in the performance** of their professional duties.

Professional Society Guidance (cont)

ASME Fundamental Canons:

9. Engineers **shall not seek ethical sanction against another engineer** unless there is good reason to do so under the relevant codes, policies and procedures governing that engineer's ethical conduct.

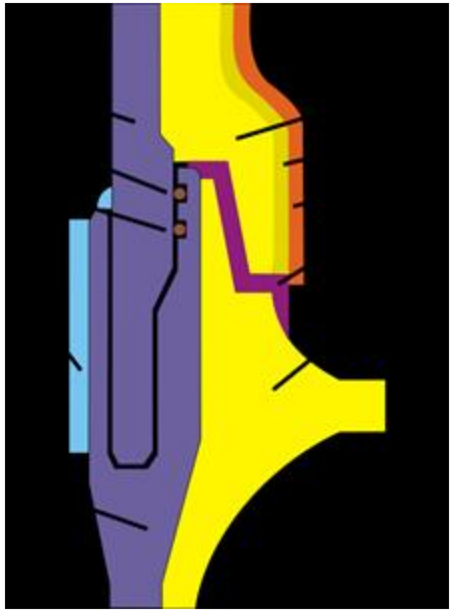
10. Engineers who are members of the Society **shall endeavor to abide by the Constitution**, By-Laws and Policies of the Society, and they shall disclose knowledge of any matter involving another member's alleged violation of this Code of Ethics or the Society's Conflicts of Interest Policy in a prompt, complete and truthful manner to the chair of the Ethics Committee.

Loss of Space Shuttle *Challenger* (Jan 1986)

Challenger broke apart 73 seconds into flight resulting in the death of seven crew members (including Christa McAuliffe, who was supposed to be the first Teacher in Space).

- O-ring failure on Solid Rocket Booster (SRB) allowed pressurized hot exhaust gases to escape and led to failure of the SRB attachment and structural failure of the external fuel tank. Aerodynamic forces then caused breakup of the orbiter.

- *This was a known issue.* Since 1977 there had been concerns about that catastrophic failure of an O-ring, particularly in cold weather.



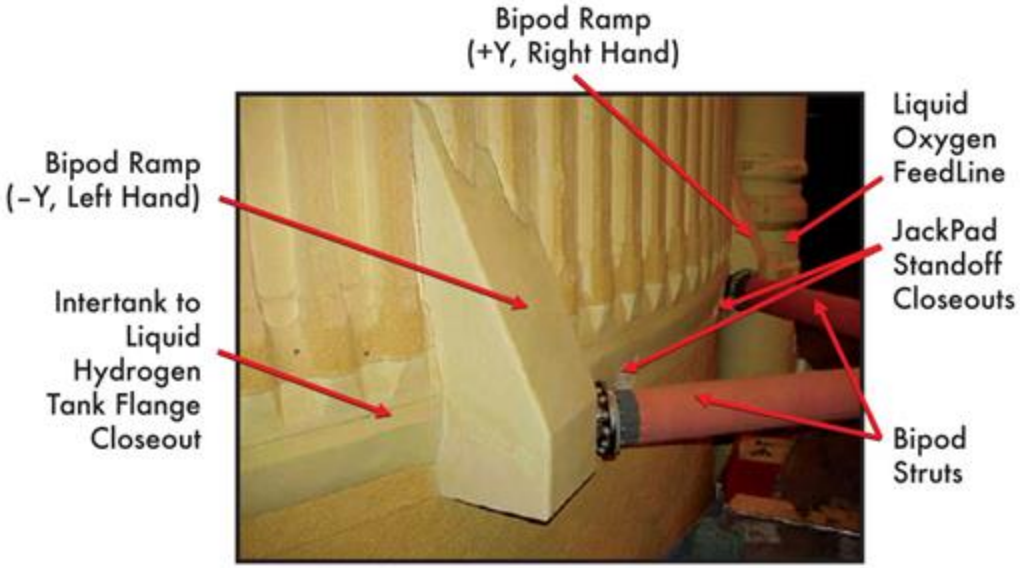
Loss of Space Shuttle *Challenger* (Jan 1986)

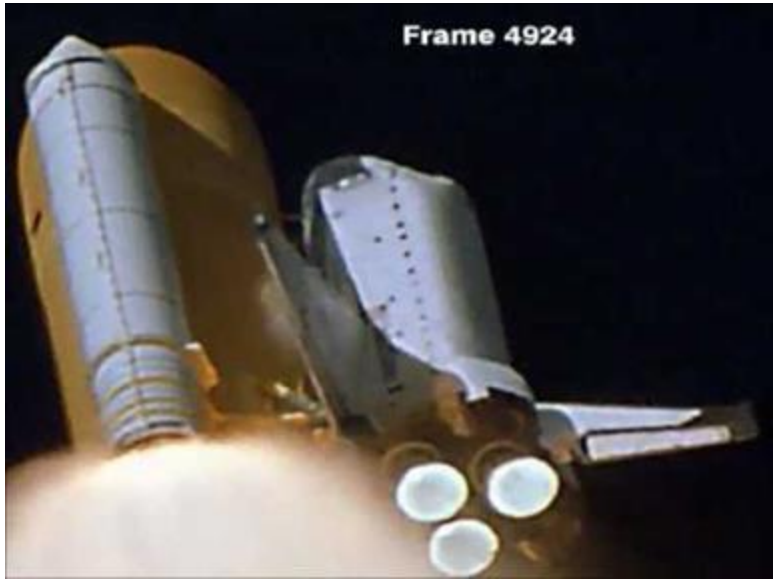
- NASA managers disregarded warnings from Morton Thiokol, manufacturer of the booster, about launching in cold weather.
- "Go Fever" as launch was delayed and NASA under significant pressure to "…make space flight as routine as airline travel…"
- O-rings "…only qualified to 40 degrees [F]...what business does anyone even have thinking about 18 degrees, we're in no man's land."
- Loss of Space Shuttle *Columbia* in Feb 2003, showed that "lessons learned" from Challenger had been forgotten….

Loss of Space Shuttle *Columbia* (Feb 2003)

- 82 sec after launch a piece of thermal insulation foam from the external fuel tank broke off and hit the *Columbia's* left wing.

- Bipod Ramp foam had been observed coming off on at least 4 prior flights.



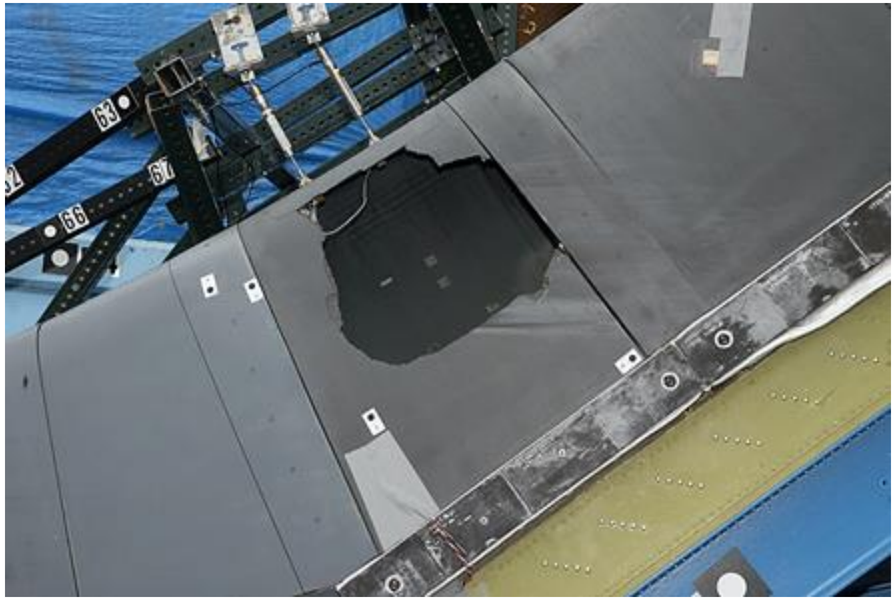


- Prior post flight safety reviews noted "…the External Tank is safe to fly with no new concerns (and no added risk) of further foam strikes."

- *Columbia* foam strike noted but no action taken ("…nothing could be done…").

Loss of Space Shuttle *Columbia* (Feb 2003)

- *Columbia* Accident Investigation Board determined that foam strike resulted in large hole in left wing (41X42 cm in ground testing).



- Hole allowed hot atmospheric gases to penetrate and destroy internal wing structure resulting in loss of the shuttle (and crew of 7 astronauts).

- Image shows left wing debris on reentry (from USAF optical telescope in New Mexico).

Loss of Space Shuttle *Columbia* (Feb 2003)

- *Columbia's* left wing failed and shuttle disintegrated @ ~ Mach 19.5 and ~ 210,000 ft altitude over east Texas.

- No chance of crew survival (bodies torn apart by aerodynamic loads).

- *Columbia* was in orbit for 15 days with no attempt to discover damage or make repairs.

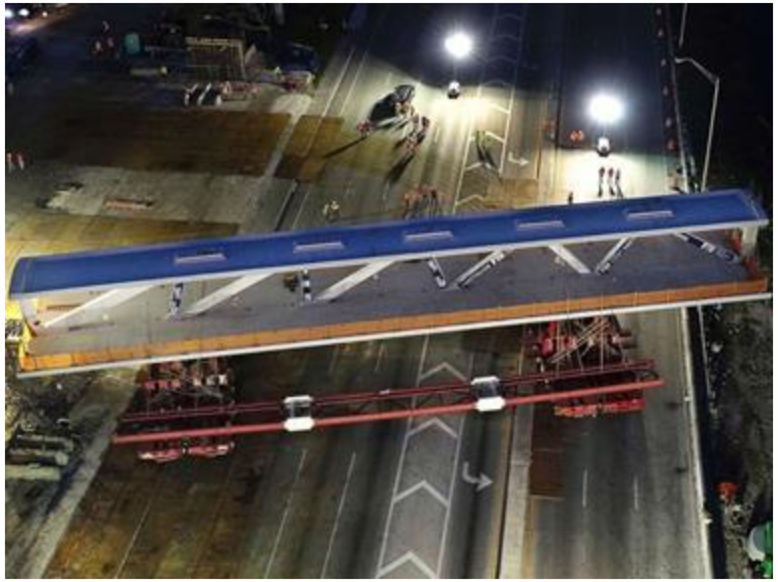
- *Atlantis* might have been able to be launched in time to rescue the crew.

- Patch to wing possible with metal from cabin?

FIU Pedestrian Bridge Collapse (15 Mar 2018)

- 320 ft long bridge collapsed during construction resulting in 6 deaths and 9 serious injuries.

- New Accelerated Bridge Construction (ABC) quick build design promoted by and supervised by Florida International University (FIU). Bridge construction normally supervised by the Florida Department of Transportation.



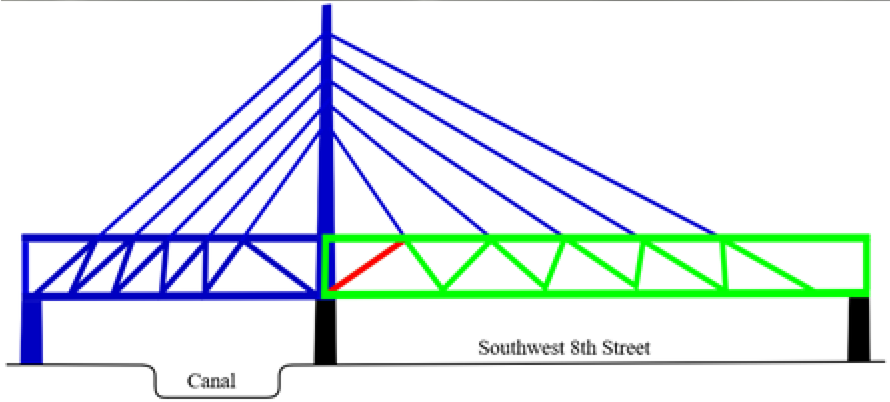
- FIU home of federally-funded ABC University Transportation Center…

- Post-tensioned concrete structure, designed to look like a steel cable-stayed bridge. Used new formulation “clean concrete” that would not get as dirty with age.

- Main structure built on the side of the road then moved into place 5 days prior to failure.

FIU Pedestrian Bridge Collapse (15 Mar 2018)

- Bridge was designed to last 100 years and survive a Cat 5 hurricane.
- Undergoing post-tensioning adjustment at the time of failure (red beam below right).
- Significant cracking observed on one north pylon (blue at left) two days prior to collapse. Voicemail left for FDOT but not accessed until the day after the accident.
- **Active NTSB investigation still in progress.**
- **Perhaps too much unproven new technology? New ABC quick build design, new concrete formulation?**
- **Pressure from FIU to have the example on campus for FIU ABC University Transportation Center?**



Ethics Assignment

- **Each group (3 students)** will create a single poster. A poster template will be provided as a guideline.
  - Groups will be assigned randomly.
  - Ethics case scenarios will be released on Blackboard.
  - Each group will be assigned one ethics case scenario.
  - Depending on the class size one of groups may have one member more or less than three members.
- **The assignment is worth 20 points, total.**
  - All students will receive the same grade unless a member(s) of the group petitions to have an uneven distribution due to large differences in contributions.
- **A rubric for grading will be provided** to ensure that your poster meets with assignment requirements.

Deliverables:

- Rough Draft (3 points) on November 23
- Final poster (5 points) on December 1
- Poster presentation (12 points) on December 3

Important points to note while doing the ethics assignment

- **Group work:**
  - Allocate work in your group appropriately
  - Assign roles and allow voice-participation for each member
  - Draft submission date is a good time to discuss with the instruction team if your poster is going as per your plan

Be respectful:
Be aware that you are working in a group. And that, you may be working with them for the first time. Be fair, mindful and empathetic to your colleagues.

Be responsible:
Help your team members. Keep an open mind to the course material presented.

Be a communicator:
Observe, Ask questions and Communicate with your team effectively and politely. Stay punctual and responsive via emails and other modes of communication.

Be a problem solver:
Explore options to complete the ethics assignment. Come prepared to your meetings. **Stay positive about the assignment outcomes.**

- **Final poster presentation**
  - Student groups should prepare for a 5-to-7-minute, timed oral presentation.
  - Student groups will be asked questions by the instruction team for about 1 minute
  - Bonus points may be awarded for well-thought responses.
  - A poster template is being provided on Blackboard.
    - Student groups must provide references for material sourced from articles, websites or other resources
    - The poster should have images that must be referenced properly unless they are original images or sketches
    - **Poster with only text will not be given full points.**

Grade Policy:

- Any questions or disputes about grading must be received within one week of having the graded work returned to the student.

- Requests must be rational and submitted in writing (email is acceptable), and the TA/professor may refuse the request.

- All of the grading for this course will be done by the GTAs. You should approach them first with any grading-related questions.

Grading:

- Lab 0, 10%
- Lab 1, 10%
- Lab 2, 10%
- Lab 3, 10%
- Lab 4, 10%
- Recitation Problems, 30%
- Ethics poster/assignment 20%

There is no option for a make-up lab for **Lab-4** dealing with fluid friction in pipes.

See the schedule posted on Blackboard and plan ahead!